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Group closeness effects on co-owned information sharing: A multilevel perspective

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ABSTRACT

Co-owned information contains personal details about multiple individuals, often nested within a social group. It is important to study the sharing of such information because its careless disclosure can violate the privacy of all co-owners. What makes such sharing decisions unique is that they are often conducted within a tight social context, the attributes of which can systematically affect the decisions of all individuals nested within the group. This necessitates multi-level theorizing and testing. Doing so, we theorize the impact of *group closeness* (a group-level attribute) on co-owned information sharing by the group members (individual-level reflections and behaviors). We tested our ideas through a deceptive procedure: ninety participants in 40 groups were asked to voluntarily share a co-owned photo of 2–3 group members, for algorithm training purposes (cover story). Hierarchical Linear Modeling revealed (1) the retained relevance of self-centered private information sharing motivators and deterrents in group contexts, and (2) a cross-level effect of group closeness: it weakened the negative effect of privacy concerns on actual co-owned information sharing. The findings underscore the role of social context in determining the potency of privacy concerns to drive the privacy behaviors of individuals nested within this context.

1. Introduction

Co-owned information (e.g., a screenshot of group chat, a group photo, or a shared document) captures jointly created, possessed, and controlled information by multiple individuals (Belanger & James, 2020; Petronio, 2010). These individuals can harbor varying degrees of ownership, which is reflected in the extent to which they claim rights to and exert control over the shared information (Belanger & James, 2020; Petronio, 2010). Those with at least some ownership perceptions are information co-owners (Petronio, 2010). Because co-owned information captures personal details or data that pertain to multiple co-owners, the intentional or unintentional sharing of such information might pose risks, such as privacy breaches, the straining of good relationships, and embarrassment (Feng et al., 2024; Li & Gui, 2022; Such et al., 2017; Tierney et al., 2013) to all co-owners. Such risks are exacerbated with modern technologies, such as algorithmic inferential analysis (Taylor et al., 2016). This is because individuals' preferences and behaviors can be inferred through their co-owned information with others (Mittelstadt, 2017). Therefore, studying co-owned information sharing has important implications for effective privacy protection in today's surveillance economy (Clarke, 2019).

Current information systems (IS) research has primarily studied individual, self-centered sharing behaviors, as explained by the APCO (Antecedents → Privacy Concerns → Outcomes) model and its extensions (Dinev & Hart, 2006; Dinev et al., 2015; Xu et al., 2009). This model assumes that information sharing decisions depend on the balance between motivating factors and safeguards (e.g., benefits one has from sharing and trust) and sharing deterrents (e.g., privacy risks, privacy concerns) (Buck et al., 2022; Wisniewski & Page, 2022). However, in the case of co-owned information (e.g., a joint photo), the focal individual who makes sharing decisions, as well as other co-owners (e.g., other individuals in the picture) might all be affected by the sharing behavior. This can shift the focal individual's considerations from being self-focused to becoming at least somewhat guided by properties of the social context (group) (Belanger & James, 2020).

Hence, there is a need to integrate social context attributes with privacy decisions (Xu & Zhang, 2022), and to better theorize the context-dependent sharing of co-owned information. Social context captures the surrounding associated with an object or phenomenon, often manifested in higher-level units of analysis (e.g., groups) (Johns,

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2006). Hence, in the case of co-owned information sharing, the group of co-owners is a well-defined social context. This context can have various attributes that vary between groups (e.g., number of members, the tenure of the group, and how tight-together group members are). Such group-level attributes are unique in that they simultaneously and systematically influence all individuals nested within the group, which makes people nested within a group think and react in interdependent ways (Kozlowski et al., 2013). Thus, through overarching influences, group-level factors can systematically alter the processing and weighing of individual-level considerations and behaviors by people nested within groups (Klein et al., 1999). As such, understanding phenomena such as individual behavior within a social context requires multi-level theorizing and testing that takes into account both individual-level and group context variables (Kozlowski et al., 2013).

Notably, several studies have started examining the sharing of co-owned information (see Appendix). This body of research provides valuable insights into how factors such as psychological ownership (Feng et al., 2024) and the mental weighing of information sharing deterrents vs. motivators influence co-owned information disclosure on social media (Feng et al., 2024, 2023; Kim & Gweon, 2016). However, despite the importance of such works, they have largely emphasized individual-level reflections such as the privacy risks associated with shared information use (Li & Gui, 2022), the extent to which the privacy of co-owners could be violated (Such et al., 2017), and factors that influence peer privacy concerns (Zhang et al., 2022). Nevertheless, they have overlooked possible group-level influences. Addressing this gap is important as it would allow a more comprehensive and nuanced understanding of co-owned information sharing decisions (Belanger & James, 2020).

Here, we address this gap by building on Johns (2006)'s theory of context impact and applying it to the case of co-owned information sharing. According to Johns (2006), while groups can have many attributes, the tie and strength of interpersonal relationships among group members represent a key group attribute that affects the way group members think and behave: it shifts emphasis from self-focused to more group-focused thinking. Applied here, we posit that people's strength of relationships with others in the group is one important group attribute that can also alter the way people in the group process privacy considerations and make privacy decisions (Kim & Gweon, 2016; Such et al., 2017). Hence, we theorize and measure group closeness, defined as the groups' overall tie strength, psychological connection, and cohesion among group members (Ellemers et al., 1997). Viewed as a group-level attribute, group closeness can manifest cross-level effects on individuals' privacy decision-making, making it highly relevant to our multilevel approach. In addition, it is a universally common group attribute, regardless of group types (e.g., work groups or friend groups). As such, studying it can provide practical insights into co-owned information sharing in groups with different levels of tie strengths, such as parent-child groups (Peng, 2023) and social media groups (Utz & Beukeboom, 2011). All in all, we focus here on this potentially important group-level factor, while acknowledging that this is certainly not the only or always the most important factor at the group-level. We therefore seek to address the following research question (RQ):

RQ. : *How does group closeness (a contextual [group-level] factor) impact the sharing behavior of co-owned information by the individual group members?*

To address this RQ, we theorize that group closeness can influence the sharing behavior of co-owned information via two mechanisms. First, it can attenuate the weight people give to their own privacy concerns in decision making. This happens because interpersonal closeness affords social cognition about other group members (Erle & Topolinski, 2017; Ho & Ng, 2022), which is less motivated in groups with weak ties (Peterson et al., 2015). Hence, in high group-closeness conditions, we can expect that individuals might exhibit more concerns towards others and consequently lower their self-focused emphasis in disclosure decisions. Second, group closeness can also directly impact sharing behavior, because it creates an atmosphere of psychological safety (Jowett et al., 2023). When people feel safe in a group, they engage in more approach behaviors (Appelbaum et al., 2020). As such, we expect that in close groups, group members will share co-owned information more so than in groups low in closeness.

We tested these ideas using a deceptive procedure, which allowed us to measure actual sharing behavior. Specifically, a real local Machine Learning Lab asked co-owners of a group photo if they were willing to provide the co-owned photo for a machine learning project (cover story). Hierarchical linear modeling (HLM) techniques applied to a sample of 90 people nested within 40 groups (i.e., each group contained 2–3 members) revealed that group closeness had a cross-level effect: it *weakened* (positively moderated) the negative path between individuals' privacy concerns and the actual sharing of co-owned information. However, group closeness did not directly impact actual sharing behaviors. These results contribute significantly to theory in at least three ways. First, they show that co-owned information sharing decisions still activate self-centered reflections that motivate and/or deter sharing. Second, the findings represent initial steps in multi-level theorizing and empirically examining the privacy decision-making of individuals nested within groups. They show that between-groups variation in contextual attributes can affect within-group processing of privacy decisions. Lastly, the findings show that the commonly examined effects of privacy concerns on privacy behaviors are context dependent. This highlights the need to consider additional group-level factors that can attenuate the ways in which individuals within a context process privacy reflections and act on them.

2. Related works and theoretical background

Our study is grounded in privacy decision theories (Dinev et al., 2015; Smith et al., 2011) and the theory of context impact (Johns, 2006). Hence, we elaborate on these theories, position our work within existing research, and explain the need for a multilevel perspective to study co-owned information sharing.

2.1. Private information disclosure

Private information disclosure captures the voluntary sharing of personal information with others. It is a common behavior that takes place in settings such as social media (Sun et al., 2015), e-commerce (Dinev & Hart, 2006), online health communities (Li et al., 2024), and human-AI interactions (Saffarizadeh et al., 2024). Such behaviors are often explained by the APCO (Antecedents → Privacy Concerns → Outcomes) model and its extensions (Buck et al., 2022; Wisniewski & Page, 2022). Overarchingly, the APCO suggests that people decide on privacy behaviors by contrasting factors that motivate sharing or make it more palatable (e.g., benefits or trust) against factors that demotivate sharing (e.g., privacy concerns and privacy risks) (Belanger & James, 2020; Dinev & Hart, 2006; Dinev et al., 2015; Smith et al., 2011). Extending this overarching view, studies have shown that private information sharing can be influenced by additional reflections, such as data ownership (Cichy et al., 2021), mood states (Alashoor et al., 2023), social cues (Zalmanson et al., 2022), altruistic values and privacy norms (Li et al., 2024).

Importantly, the APCO points to two possible, yet different, factors that represent deterrents (negative predictors) of information sharing: privacy risks and privacy concerns (Dinev & Hart, 2006; Dinev et al., 2015; Smith et al., 2011). Privacy concerns capture people's concerns about future privacy violations as a result of the information they disclosed, voluntarily or not (Buck et al., 2022). In contrast, privacy risks capture individuals' perceptions of the possible negative consequences as a result of what happens to their data (Dinev & Hart, 2006). Recent studies have shown that privacy risks are comparatively more unpredictable and ambiguous in an online environment because of privacy uncertainty (Acquisti et al., 2015). Moreover, privacy risks are sometimes too complex to fully predict because of their multidimensional nature (Karwatzki et al., 2022). In contrast, privacy concerns are based on more stable privacy experiences and levels of awareness, as influenced by individual differences (Buck et al., 2022). Thus, in this study we focus on privacy concerns as the deterring factor in one's sharing decisions, without discounting the possible importance of privacy risks.

To contrast privacy concerns, we focus here on the perceived benefits of sharing as a key motivator of sharing. We need such a factor because privacy behavior models assume such weighing of motivators and deterrents as the mechanism that helps people decide whether to share (or not) private information (Wisniewski & Page, 2022). Privacy benefits can include utilitarian benefits (e.g., customized services and products) (Xu et al., 2009), hedonic benefits (e.g., enjoyment and fun) (Sun et al., 2015), and social benefits (e.g., relationship building) (Cheung et al., 2015). Prior research suggests that hedonic benefits are more immediate and proximal because online information sharing is intrinsically rewarding (Tamir & Mitchell, 2012).

Here, we focus on hedonic benefits (pleasure gains from providing data) as the motivating factor for two reasons, without discounting the potential importance of non-hedonic benefits or other sharing motivators. First, there is psychological and biological evidence that the mere sharing of information is enjoyable/intrinsically rewarding; this happens, because keeping information/secrets is burdensome and humans have an innate desire to share information (Tamir & Mitchell, 2012). Indeed, prior studies consistently demonstrate that hedonic value exerts a particularly strong influence on voluntary information sharing behaviors, often surpassing utilitarian considerations. For example, Krasnova et al. (2012) emphasizes that enjoyment significantly motivates users to disclose information on social networking sites. Sun et al. (2015) also reveals that hedonic benefits could significantly motivate online information sharing, in tandem with utilitarian benefits. Extending this idea to the case of co-owned information, Feng et al. (2024) also alludes to the effect of hedonic benefits on the sharing of co-owned information. All in all, while other benefits (e.g., utilitarian and social) might be important, there is at least some reason to believe that hedonic benefits motivate sharing behaviors (Tamir & Mitchell, 2012).

Second, for convenience reasons, we focused on a motivator that is intrinsic and does not require cultivation over time. This choice was driven by our desire to be able to employ a study design that does not require an ongoing interaction with the online entity. While people can sense the enjoyment of sharing even in one instance (Tamir & Mitchell, 2012), we expected other important motivating factors, such as trust (beliefs about the benevolence, integratory and competency of the website and its users (Dinev & Hart, 2006)), to have limited meaning beyond gut-feelings, when individuals encounter a one-time unfamiliar online vendor. Such situations are common in life (e.g., sharing information with a market research firm one has never heard of) and in experiments (Li et al., 2011, 2008). Moreover, factors like trust can be intertwined with privacy concerns as they have co-influences (Buck et al., 2022), which makes their separation less practical.

Taken together, these arguments suggest that (1) information sharing behavior may be driven, at least in part, by hedonic benefits, and (2) while such a focus is not overarching, it allows for testing in experimental settings that involve swift interactions with an unknown online entity. By affording such designs, we also extend prior research that has primarily focused on sharing intentions, and examine whether hedonic benefits also motivates actual sharing behavior.

Notably, focusing on privacy concerns as the sharing deterrent, and contrasting it with perceived benefits, as the sharing motivator, is a partial view of the full APCO (Buck et al., 2022). However, given the abovementioned reasons, we have chosen to take this limited view, as it is a reasonable starting point for understanding how people balance sharing deterrents and motivators in privacy decisions. This also follows from prior studies that did not test all deterrents and motivators, and instead, focused on the contrast between perceived benefits and privacy concerns (Breward et al., 2017; Hall et al., 2024). Here, we suggest that while this limited view (and perhaps more broadly, the full APCO) is a good start, it needs to be extended to explain the sharing of co-owned information. This is because this view (or the full APCO) emphasizes self-focused individual-level reflections (e.g., of benefits and concerns), but overlooks the possible role of social context in privacy decisions. Thus, we seek to extend this literature stream by theorizing and testing context effects on the self-focused perspective.

2.2. Co-owned information disclosure

Multiple individuals can co-generate, co-regulate, and co-share private information (Petronio, 2010). Hence, privacy decisions can also commonly manifest within groups (Bélanger & Crossler, 2011; Belanger & James, 2020). Co-owned information sharing is an example of this, as it manifests in situations where multiple individuals own the content and can be impacted by sharing decisions.

Co-owned information arises in two key situations: (1) when multiple individuals collaboratively create content that reflects the personal details of all involved, such as a group photo, or (2) when one person has access to and control over another person's personal information, for instance a selfie of a friend (Petronio, 2010). The latter type of sharing is often termed peer disclosure (Cao et al., 2018; Zhang et al., 2022), interdependent information disclosure (Alsarkal et al., 2018), or disclosure of others' information (Koohikamali et al., 2017). While this stream of research is important, we specifically focus on the first type, for the following two reasons. First, this type of co-owned information disclosure is unique because the shared information simultaneously captures personal details of the focal individual (who makes the sharing decisions) and other co-owners. This distinct aspect differentiates it from self-disclosure and peer disclosure. It directly exposes the personal details of all co-owners and thus exacerbates privacy threats to all involved. Second, it transcends the conventional self-focused viewpoint as individuals' decision-making processes can be subtly guided by their contextual considerations (i.e., group-level influences). In this sense, the sharing of co-owned information of individuals nested within groups is unique, as it involves entities in at least two levels of theorizing and analysis: (1) groups, and (2) individuals within these groups.

Prior research has started exploring co-owned information disclosure through the lens of the APCO model (Feng et al., 2024, 2023; Jia & Baumer, 2022; Kim & Gweon, 2016). While in its nascent stage, this emerging body of work offers two key insights. First, it shows that the typical weighing of self-centered information sharing deterrents vs. motivators from the APCO model impacts co-owned information disclosure (Feng et al., 2024, 2023). This means that individuals still partially rely on APCO components, such as perceived self-centered benefits and concerns, when deciding whether to share information that involves others. Second, the APCO model alone does not fully capture the complexities of co-owned information disclosure. Extensions are needed to incorporate additional factors. Nascent research has found that factors such as psychological ownership (Kim & Gweon, 2016) and privacy boundary management (Jia & Baumer, 2022) also play important roles in shaping disclosure decisions, complementing the original APCO framework.

Despite these important insights, this literature stream has not systematically theorized possible effects of group attributes on the sharing decisions of individuals nested within groups, despite the fact that such group (context) influences are powerful and important (Johns, 2006). This is in line with ideas that privacy decision making is context dependent (Nissenbaum, 2004). Thus, it is crucial to consider the impacts of contexts (i.e., group-level influences) on the co-owned information privacy decisions of group members, where groups of co-owners represent a well-defined context (Jones & Duncan, 1998; Kozlowski et al., 2013).

By contrast, our study builds on and extends this prior research in two meaningful ways. First, we draw on the APCO model by proposing that self-centered sharing deterrents and motivators can still influence co-owned information disclosure. By doing so, we validate the model's relevance in this more complex and socially embedded context, thus strengthening our theoretical foundation. Second, our unique contribution lies in extending the sharing deterrents and motivators from the APCO model to the group level. To this end, we theorize how individual privacy considerations (concerns vs. benefits) are systematically shaped by group-level attributes and apply multilevel analytical tools to test these ideas. This represents a meaningful advancement, as prior work has largely focused on individual-level APCO components, even when investigating the sharing of socially embedded information, thus overlooking the broader social and contextual influences that our study brings into focus.

2.3. Group closeness

To study group-level influences on co-owned information sharing, we focus here on group closeness, which captures the sense of psychological connection, bond, and cohesion among group members (Ellemers et al., 1997). We do so for three reasons. First, when information is co-owned and the sharing of such information can impact all co-owners, individuals' sharing decisions may be affected by attributes of their bonds with other co-owners (Feng et al., 2024, 2023; Jia & Baumer, 2022; Kim & Gweon, 2016). This is because group closeness, as a manifestation of bond strength, can evoke group identity (Brown, 2000) and psychological safety (Coutifaris & Grant, 2022). Accordingly, bonding strength can drive individual members to think and act in interdependent ways (Belanger & James, 2020). Applied here, when making privacy decisions that might have impacts on others, individual members nested in groups may process privacy considerations differently, depending on their bond strength with other intra-group members. Thus, group closeness is an important contextual factor that can impact the sharing behavior of individuals within the group. Second, research in privacy-sensitive contexts (e.g., IoT settings) consistently shows that social cues from close others exert a stronger influence on individuals' behavior than cues from distant or expert sources. This suggests that closeness influences the weight individuals assign to their own evaluations of privacy risks and consequences in privacy decisions. In particular, denials or caution from close others act as strong normative signals, prompting more protective behavior (e.g., (Emami Naeini et al., 2018)). Third, this behavioral pattern aligns with established theoretical models of privacy regulation. Both Altman's Privacy Regulation Theory and Communication Privacy Management Theory conceptualize privacy as a relationally negotiated and context-dependent process. In these models, interpersonal closeness plays a critical role: it facilitates the negotiation of shared privacy boundaries and alters the strategies people use to manage disclosure. Strong relational ties are theorized to foster implicit trust and smoother coordination in privacy management, making closeness not just a correlate but a mechanism that underpins joint privacy behavior (Altman, 1975; Petronio, 2010). Fourth, group closeness is relevant across group types (e.g., work teams, student groups, and family groups). This enhances the potential relevance of the concept, beyond the current study. Last, co-owned information sharing often occurs in close groups, for instance "sharenting" by parents (Peng, 2023). Thus, understanding how this context attribute can affect co-owned information sharing could also provide practical implications for prevalent phenomena.

Ultimately, these insights underscore group closeness as both a theoretically grounded and empirically supported construct for understanding how individuals navigate the complexities of co-owned information sharing. While other group-level constructs such as identity or belongingness are undoubtedly relevant, closeness offers a psychologically immediate and interactionally salient lens through which privacy decisions are experienced and enacted in groups. Hence, we believe studying group closeness represents a good starting point towards multilevel privacy research by examining its cross-level effects. We acknowledge that this is not the only or always the most important of factor; there can be many other group-level constructs involved in privacy decisions. We simply posit that group closeness is one important factor, but examining additional factors is an important future research direction (see Section 6).

The idea of context effects on individuals nested within the context is based on Johns (2006)'s theory of context impact. Social context is the surrounding of a phenomenon, which usually captures factors associated with higher-level units of analysis (e.g., groups) (Johns, 2006). For example, organizational characteristics can provide contexts for individual members and external environment can provide contexts for organizations (Johns, 2006). This happens because when individuals are nested in a hierarchical social structure, their psychological processes, decisions, and behaviors can be influenced by properties of this higher-order social structure, and example of which is group of information co-owner (Bélanger et al., 2014). Consequently, decision-making processes of individuals within a social structure can be affected by social (higher-order) structure properties; this reflects a cross-level effect (Molina-Azorín et al., 2020).

Johns (2006)'s theory was initially designed for organizational studies, particularly centering on how team-level characteristics (e.g., team cohesion) can affect individual members' work performance (Magni et al., 2009). Recent IS research has started using Johns (2006)'s ideas to explain context, albeit not specially group, effects on privacy decisions (Bansal et al., 2016; Xu & Zhang, 2022). Here, we adapt such ideas to co-owned information sharing by individuals nested in groups. Specifically, we conceptualize group closeness as a group-level attribute (i.e., a contextual factor), and expect it to have a cross-level effect on privacy decisions by individual team members. Building on this notion, we develop hypotheses in the following section.

3. Hypotheses

Our theory relies on the idea that people within a group still follow the self-centered reflections on sharing deterrents and motivators (H1-H2). However, they are also exposed to group-level influences, manifesting in the alteration of the weights they give to their own privacy concerns (H3), and direct effects on the sharing behavior (H4). The proposed multilevel model is depicted in Fig. 1.

3.1. Effect of self-centered reflections on sharing co-owned information

The APCO model suggests that privacy behaviors are negatively influenced by one's sharing deterrents, such as privacy concerns, and positively influenced by sharing motivators, such as benefits of sharing (Belanger & James, 2020; Buck et al., 2022; Dinev & Hart, 2006; Dinev et al., 2015; Smith et al., 2011). This perspective has been supported in the context of sharing personal information that is self-owned, for instance, sharing personal profiles on social media (Min & Kim, 2015), as well as in the context of co-owned information (Feng et al., 2023; Such et al., 2017). This self-focused perspective is still relevant in the case of co-owned information, because co-owned information not only exposes other co-owners but also reveals information about the focal individual (Feng et al., 2024). The theory of multilevel information privacy (Belanger & James, 2020) also echoes this view and proposes that even in complex social settings, privacy-related choices are still influenced by individuals' self-centered perceptions of benefits and concerns. This is because individuals tend to be guided, at least to some extent, by self-referential thoughts and deliberations, even when interacting in social settings (Ajzen, 1991). Therefore, it is logical to surmise that self-focused sharing deterrents and motivators extend to the case of co-owned information. In doing so, we systematically build on and extend existing literature by demonstrating how individual-level self-centered privacy considerations are influenced by group-level factors, thus highlighting our cumulative contributions. Hence:

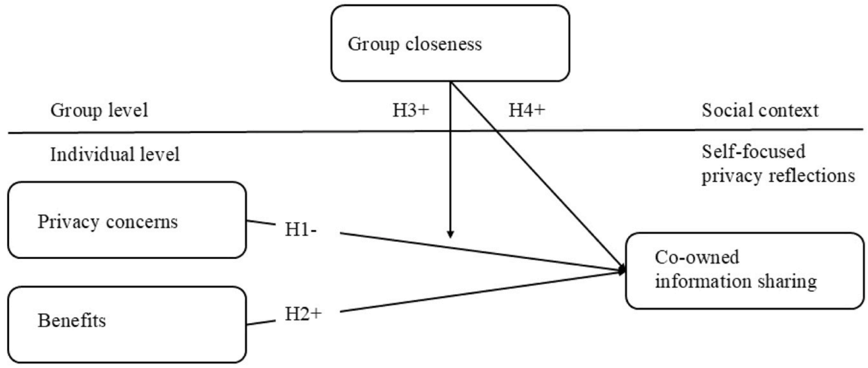

Fig. 1. Research model.

H1. : Privacy concerns of individual group members will decrease the likelihood of them sharing co-owned information.

H2. : Perception of benefits of individual group members will increase the likelihood of them sharing co-owned information.

3.2. The moderating role of group closeness

The APCO model has been extended to account for heuristics and contextual forces that can change the way people reflect on sharing deterrents and motivators (Alashoor et al., 2023; Dinev et al., 2015; Kehr et al., 2015; Li et al., 2024). Such extensions are needed because people often adopt mental shortcuts and use peripheral cues to make decisions (Acquisti, 2009; Acquisti et al., 2015; Dinev et al., 2015).

Here, we focus on the novel role of group closeness, a group-level attribute, in altering the weighing of privacy concerns by people nested within these groups. Group closeness is a contextual factor, which may subtly shift individuals' privacy-related reflections from being self-focused to being more guided by the broader group context. In close-knit groups, members are typically more socially attuned, meaning that they tend to consider others' needs and reflect on the interpersonal implications when their own decisions might have impacts on other group members (Kawakami et al., 2017). This illustrates how a sense of closeness can trigger social cognition (i.e., a cognitive ability to understand and empathize with other people), such that individuals focus more on others rather than solely on themselves (Khabbache et al., 2020).

Importantly, when making decisions for others, individuals tend to pay greater attention to the potential negative outcomes that might happen to other people, compared to when making decisions for themselves (Polman & Wu, 2020; Wray & Stone, 2005). This tendency is explained by the nonmaleficence moral principle of viewing placing others in an unfavorable situation as morally wrong (Cushman et al., 2006). When people apply this logic, the focal individual's attention shifts away from personal concerns because of a heightened focus on the wellbeing of others. Given the heightened focus on negative implications for others (Polman & Wu, 2020; Wray & Stone, 2005), this effect should be pronounced for sharing deterrents, such as concerns. This also means that there is no reason to expect changes in sharing motivators (e.g., benefit considerations), because such reflections do not tend to change in close-knit groups. Prior research has also shown that hedonic benefits are more self-referential (Sun et al., 2015; Xu et al., 2025), making them less dependent on group-level variation. Hence, we posit that group closeness will weaken the negative effect of self-centered privacy concerns on sharing co-owned information (i.e., it will make the effect more positive). In highly bonded groups, individuals will become less self-centered and emphasize the group in their privacy decisions. In contrast, individuals in groups low in closeness will likely emphasize their own concerns rather than engage in holistic reflections on the group and its members. Thus, we can expect that:

H3. : In groups high in group closeness, group members will present a weaker (less negative) effect of privacy concerns on sharing co-owned information.

3.3. Direct effect of group closeness on co-owned information sharing

Prior studies suggest that individuals feel safer in close groups (Porter et al., 2024; Sanner & Bunderson, 2015). In such cases, people have a sense of false safety, as they assume no harm to them if they make a mistake (Jowett et al., 2023). They tend to have a more open attitude toward information sharing, such as freely voicing opinions and expressing ideas publicly (Oztop et al., 2018). These advantages of psychological safety in close groups also mean that people in such contexts often engage in more approach behaviors (Jowett et al., 2023; Polman & Wu, 2020), compared to low-closeness groups. Using the above ideas, we can speculate that, regardless of one's privacy concerns, members in close groups may share co-owned information more than the members in groups low in closeness, given their false sense of psychological safety within the group (Prinstein & Wang, 2005). As such, we expect that in close groups, group members will share co-owned information more so than in groups low in closeness:

H4. : In groups high in group closeness, group members will present a higher likelihood of sharing co-owned information.

4. Methodology

4.1. Cover story

We tested our hypotheses with a deceptive paradigm, in which we used a reliable cover story so that we can observe participants' *actual* co-owned information sharing behaviors. The cover story was that the researchers worked on an image recognition software that could classify pictures based on machine learning techniques. Since this software was still in development, we needed to recruit participants who were willing to share a group photo (including themselves) to help optimize the algorithms. We did not reveal the true research purposes during recruitment. Simply put, participants were led to believe that their data would be used and processed by machine learning algorithms.

This deceptive paradigm follows from prior IS research which used it for observing *actual* disclosure behaviors. For example, Xu et al. (2009) used a cover story to study self-disclosure, in which participants were led to believe that they were evaluating a location-based social networking site. Zalmanson et al. (2022) created a simulated video-sharing platform and invited participants to evaluate its user experience, where they asked participants to share their actual information (e.g., birth date). Keith et al. (2013) recruited participants under the misleading pretense that they were needed to evaluate a new mobile application, and through this approach, participants' actual disclosure behaviors were observed. In such cases, like in the current study, deception is necessary to (1) make the study realistic, (2) allow observing actual sharing behaviors, and (3) reduce the chances of participants altering their behavior in response to knowing the true purpose of the study (Junco, 2013). Like in other studies using the deception paradigm, we debriefed participants on the true purpose of the study after it was done. This procedure was approved by the local ethics committee.

4.2. Procedure

As a first step, we contacted potential participants (university colleagues and students) through the researchers' personal network. We asked them (1) if they had a group photo that included 2–3 individuals who directly faced the camera, and that they have never posted online; (2) what types of relationships were among these individuals. To encourage participation, a raffle of three Amazon vouchers of 50 Euros for each participant was provided. We let them know that participation in the raffle does not depend on whether they share the photo with us.

At this first stage, we ensured that (1) these group photos were similar in the number of people captured (2–3 individuals), the physical surroundings or environment depicted (either a restaurant or nature), the activities shown (social gathering), and the information valence (neutral or positive); (2) potential participants received the same cover story and instructions; (3) participants were nested within diverse types of social groups with varying levels of relationship strength. We identified 232 potential survey participants who belong to various groups, each captured in a distinct co-owned photo. Formal survey invitations were distributed via email to these individuals by a local Machine Learning Lab. Each group (and its co-owned photo) was given a unique identifier, allowing us to connect group members, with their unique identifiers, to groups.

The survey began with introducing the cover story of this study, rendering participants believe that their co-owned photo sharing was needed for a machine learning project. We briefly mentioned how promising the project was, as well as acknowledged potential issues regarding data security. Then, we asked participants if they were willing to upload the group photo identified in the first stage, to allow for machine learning training. We explicitly told our participants that uploading the picture was completely voluntary and that it will not affect their compensation. Their actual sharing behavior (the dependent binary variable) was measured by coding their actions (sharing=1, not sharing =0). Last, we asked participants to complete the survey without contacting other members shown in the picture. In cases where participants indicated coordination with others (self-reported at the end of the survey), those groups were subsequently excluded from the data analysis to maintain the integrity of the individual responses.

4.3. Measurement

Our scales were largely based on well-established instruments which were adapted to our paradigm. Privacy concerns were measured based on the scale in Dinev and Hart (2006). Benefits were measured based on the scale by Voss et al. (2003). Notably, despite the complex structure of perceived benefits (e.g., social benefits, hedonic benefits, and utilitarian benefits), we specifically focused on the hedonic benefits of information sharing. We did so for two key reasons. First, hedonic benefits align more with our study design while utilitarian and social benefits are comparatively not realistic in our case. In our study, which involves disclosing co-owned information in a setting without real functional gains (e.g., no product improvement, no clear social payoff), hedonic value becomes especially relevant, because information sharing itself is intrinsically rewarding (Tamir & Mitchell, 2012). Participants engage with the cover story in a way that is minimally consequential to their future actual benefits, making immediate hedonic value a likely psychological driver. Second, individuals may derive intrinsic satisfaction from the idea of contributing to an innovative technological process that supports scientific progress. Participating in such a context can evoke feelings of excitement, relevance, or pride, stemming from the perception of "being part of" something meaningful and future-oriented. Group closeness was measured at the individual level based on the scale in Ellemers et al. (1997) and was then aggregated to the group level using common aggregation checks and procedures (Chan, 1998). Table 1

4.4. Sample

Out of the 232 invitees, 106 completed the study (46 % response rate). We deleted 16 participants due to failed attention checks. The final sample consisted of 90 participants (60 % females) nested within 40 groups (with 2–3 members per group). Our sample size adheres to the multilevel modeling guidelines (Hox & Kreft, 1994; Kozlowski et al., 2013; Sagan, 2019). The age distribution was: 18–24 years old (6.7 %), 25–29 years old (30 %), 30–34 years old (28.9 %), 35–39 years old (13.3 %), and over 40 years old (10 %).

5. Data analysis and results

Because the correlations between our marker variable, namely the blue attitude (Simmering et al., 2015) and other variables (-0.027–0.094) were small, we concluded that the risk for common method bias was negligible.

5.1. Individual-level constructs checks

We used SPSS 27 and HLM8 to estimate the cross-level model.

Using the individual-level responses, we calculated the reliability, convergent validity, and discriminant validity (heterotrait-monotrait, HTMT ratio) of constructs. After dropping item #1 from group closeness, all factor loadings exceeded a threshold value of 0.7, Cronbach's alphas were over 0.7, measures of internal consistency were over 0.7, and measures of convergent validity (AVE) were over 0.5 (see Table 2). Hence, the data demonstrated sufficient convergent validity.

The construct cross-loadings provided evidence of discriminant validity as assessed through the HTMT of correlations (Hair et al., 2011; Henseler et al., 2015); see Table 3. All HTMT ratios were < 0.85, which suggests sufficient discriminant validity (Benitez et al., 2020).

After these preliminary checks, a measurement model was specified and estimated using AMOS 27.0. Confirmatory factor analysis indicated that this model fits the data well [$\chi^2(40) = 50.918$ (non-significant, $p < 0.116$), CFI $= 0.986$, IFI $= 0.986$, RMSEA $= 0.055$ with p-close $= 0.399$, and Standardized RMR $= 0.0593$].

5.2. Group-level construct checks

We next tested whether within-group agreement regarding "group closeness" is sufficient as each group member answered the survey separately. This concept has a meaning only when all group members reasonably agree about the level of within-group group closeness (Chan, 1998; James et al., 1984). Specifically, we assessed within-group agreement (r_{wg}) following the approach by James et al. (1984). It revealed that the values for each group (0.727–1.000; average = 0.930) are greater than the threshold of 0.700. This indicated that individuals who belong to the same group provided reasonably similar assessments of group closeness and had a shared understanding of the level of group closeness in their group. Because we found acceptable reliability at the individual level by Cronbach's alpha, and of the group-level construct by revealing acceptable within-group reliability scores, individual-level and group-level constructs were operationalized by the within-individual and within-group means, respectively (Chan, 1998).

5.3. Hypotheses testing

We estimated the model with an incremental approach using HLM8. First, we estimated the null model to test whether the variance between groups in the dependent variable is different from zero, i.e., that there is sufficient between-group variation. A Chi-Squared test ($\chi^2 = 56.68$, df $= 39$, $p = 0.033$) indicated sufficient between-group variation in sharing and supported the need for HLM.

In the second model, we include level-1 predictors, privacy concerns and benefits, to test H1-H2. The results show a negative effect of privacy concerns on sharing behavior (-0.888, $p < 0.001$), and a positive effect of benefits on sharing behavior (1.081, $p < 0.05$). Adding the control variables, gender, and age, did not change the results.

In the third model, two-level effects were specified and estimated. In this model, the self-focused privacy reflection effects at the individual level (H1-H2) remained significant. There was support for H3: group closeness weakened (0.695, $p < 0.05$) the negative effect of privacy concern on sharing. There was no support for a direct effect of group closeness on sharing behavior (H4). This suggests that group closeness operates to influence sharing behavior primarily via driving underweighing of self-focused privacy concerns. Table 4

6. Discussion

Previous research has mainly focused on individual-level self-centered privacy decision-making (Belanger & James, 2020). However, calls have been issued to also consider group-level, context effects in privacy decisions, especially when the shared information is co-owned (Belanger & James, 2020). In such situations, it is crucial to move from self-centered privacy perspectives to social-contextual considerations, because the individuals are nested within, and hence influenced by, well-defined social structures. To examine this idea, we investigated whether information co-owners still follow self-focused privacy reflections in their privacy decision making, and the different ways through which group closeness that varies between-groups can affect the actual sharing behaviors of individuals within the group. We tested the hypotheses using a deceptive paradigm that allowed us to capture group- and individual-level concepts, and to observe actual information sharing behaviors.

Our results confirm hypotheses H1 and H2, i.e., that individual privacy concerns decrease, and perceived benefits increase the likelihood of sharing co-owned information. This is not obvious. In fact, it is an important contribution, as it suggests that individual-level decision-making processes that have been traditionally applied to one's own information also apply to co-owned information in group settings. That is, people are self-centered in terms of privacy reflections, including sharing motivators and deterrents, also when the sharing is of co-owned content that can affect others. Extending this view, the paper unpacks the cross-level effects of group closeness, which reflects a shared understanding of the relationship quality within the group and can systematically influence all people nested within the group. This context (group) attribute, we posited, can (1) reduce reliance on one's own concerns in privacy decision making, and (2) elevate the sense of psychological safety (Jowett et al., 2023), which can result in riskier behaviors, such as sharing co-owned information. We show here that only one of these mechanisms was in operation. While group closeness weakened the negative effect of privacy concerns on sharing behaviors (H3), it did not directly affect sharing behaviors (H4). While group closeness did not directly increase the likelihood of sharing co-owned information (H4), it served as a buffer against the negative influence of privacy concerns (H3). This indicates that in close-knit groups, individuals may still experience privacy concerns, but those concerns are less likely to translate into restrictive sharing behavior. In other words, high group closeness appears to moderate the effects of perceived risks associated with disclosure of co-owned information, possibly because individuals in close relationships place greater trust in each other or feel a stronger sense of (might be false) shared responsibility and solidarity.

The absence of a direct effect of group closeness on sharing behavior (H4) further suggests that closeness alone is not a sufficient motivator for disclosure; its key role is in altering how strongly those concerns impact decision-making. This highlights the conditional nature of group closeness: it matters, but primarily in regulating the effects of other factors, in this case, privacy concerns.

Taken together, these results underscore the importance of considering relational context as a moderator rather than a direct driver of behavior. They also support the idea that co-owned information is managed not solely based on relationship quality, but through a more complex negotiation between emotional bonds and perceived informational risks.

Table 1
Constructs, measurement items, and their sources.

Constructs/ Variable	Definitions	Measurement	Range	Sources
(Hedonic) benefits (BE)	Individuals' pleasure gains from providing their data for ImagAIne.	1) It is fun to provide the image for ImagAIne training purposes. 2) I enjoy uploading the image for ImagAIne training purposes. 3) It is exciting to contribute the image for training purposes of ImagAIne. 4) It gives me pleasure to provide the image for training purposes of ImagAIne.	1 = Strongly disagree, 7 =Strongly agree	(Voss et al., 2003)
Privacy concerns (PC)	People's concerns about future privacy violations as a result of the information they disclosed, voluntarily or not with ImagAIne.	1) I am concerned that the picture that I upload for ImagAIne training may be misused. 2) I am concerned about sharing the picture with ImagAIne because other people might find private information. 3) I have concerns about giving the picture to ImagAIne for training purposes because others might do something unexpected with the photo. 4) I am concerned about sharing the image with ImagAIne because the photo could be used in ways I did not anticipate.	1 = Strongly disagree, 7 =Strongly agree	(Dinev & Hart, 2006)
Group closeness (GC)	The sense of psychological connection, bond, and cohesion among group members	1) We are proud of each other. 2) We feel comfortable with each other. 3) We have respect for each other. 4) We feel together as a friendship group.	1 = Strongly disagree, 7 =Strongly agree	(Ellemers et al., 1997)
Co-owned information disclosure	The actual sharing of a co-owned photo with ImagAIne.	1) Non-sharing 2) Sharing	0 = Not providing the photo, 1 = Providing the photo	Self-developed

Table 2
Descriptive statistics, construct reliability, and validity.

Descriptive statistics						**Construct reliability and validity**		
Items	M	STD	V	FL	CC	α	CR	AVE
BE1	4.51	1.664	2.770	0.973	0.879			
BE2	4.44	1.690	2.856	0.944	0.897	0.933	0.950	0.827
BE3	5.14	1.540	2.372	0.797	0.730			
BE4	4.49	1.651	2.725	0.913	0.867			
PC1	3.26	1.618	2.619	0.935	0.853			
PC2	3.23	1.642	2.698	0.982	0.819	0.938	0.947	0.818
PC3	3.59	1.741	3.031	0.884	0.900			
PC4	3.68	1.816	3.300	0.806	0.845			
GC2	6.09	1.002	1.003	0.848	0.626			
GC3	6.27	0.747	0.557	0.823	0.598	0.727	0.867	0.685
GC4	5.64	1.425	2.029	0.812	0.555			

Note: BE=benefits, PC=privacy concerns, GC=group closeness, STD=standard deviation. M = Means, STD = Standard deviation, V= Variances, FL= Factor loading, CC= Corrected item-total correlations, α=Cronbach alphas, CR= Composite reliability, AVE= Average variance extracted.

Table 3
Correlation Matrix.

Constructs	**Benefits**	**Privacy concerns**	**Group closeness**
Benefits	-		
Privacy Concerns	0.665	-	
Group Closeness	0.090	0.100	-

6.1. Theoretical contributions

The paper presents several noteworthy theoretical contributions. First, our contribution to the privacy literature lies in the affirmation of the applicability of the self-focused privacy reflections as explained by models such as APCO, to co-owned information, thereby expanding APCO's scope to within-group contexts. We show that within-group differences are explained well by self-centered privacy reflections (i.e., benefits and concerns). Such effects have been the focus of extant research, but not within groups. The difference is that in our case, benefits and concerns considerations represent differences from the mean level within the group, as opposed to simple levels of benefits and concerns considerations in individual-level research. The interpretation here should be accordingly: the more people feel benefits above, and concerns below the within-group mean level of benefits and concerns, and correspondingly, the more they are likely to share co-owned information. This extends our understanding of privacy decisions to a new context by using a different lens.

Second, we incorporate group closeness as an important group-level factor into specific relationships in the APCO model. This not only offers a novel contextual privacy decision making perspective but also unpacks how group-level factors can influence privacy decision-making. The privacy literature has focused primarily on individual-level reflections and overlooked social structure effects (Belanger & James, 2020). Since co-owned information involves multiple individuals nested within a social structure, the sharing of such information should also be driven by group properties, as per Johns (2006)'s theory of context impact. We show that group closeness is one relevant group attribute that can affect co-owned information sharing by reducing the weight group members give to privacy concerns in their privacy decisions. This observation allows a more nuanced understanding of the privacy decision process and its boundary conditions when applied to groups and to co-owned information. This also opens two research paths. The first is a call for additional group-level research focusing on group closeness effects in other contexts. For instance, other phenomena that happen within groups (e.g., algorithm aversion by a team that uses AI) may be affected by group attributes such as closeness. The second is a call for the examination of additional group attributes beyond closeness (e.g., group trust and group homophily), as boundary conditions for privacy decision models.

Third, adopting a multilevel perspective, this study contributes to the privacy literature by extending individual-level privacy theories to a multilevel one. In line with Johns (2006)'s theory of context impact, we used a multilevel perspective to understand how group-level factors affect the privacy decisions of individual group members. This was done because in many contexts (e.g., work teams, families), individuals are

Table 4
HLM results.

	Model 1		**Model 2**		**Model 3**	
	Without controls	With controls	Without controls	With controls	Without controls	With controls
Group Level Variables: Level 2						
Direct effect of group closeness (H4)					−0.292 (0.484)	−0.299 (0.480.)
Moderation of group closeness (H3)					0.695* (0.336)	0.674* (0.349)
Individual Variables: Level 1						
Gender		−0.204 (0.511)		0.321 (0.663)		0.176 (0.655)
Age		0.020 (0.115)		0.023 (0.155)		0.028 (0.154)
Benefits (H1)			1.081** (0.366)	1.080** (0.370)	1.104** (0.407)	1.100** (0.406)
Privacy concerns (H2)			−0.888*** (0.190)	−0.900*** (0.183)	−1.038*** (0.222)	−1.041*** (0.223)
# groups	40	40	40	40	40	40
# individuals	90	90	90	90	90	90

Note: ***$p < 0.001$; **$p < 0.01$ *$p < 0.05$, Standard errors in parentheses below coefficient estimates. Entries are estimations of the fixed effects with robust standard errors. All variables have been centered around the grand mean.

nested within social groups and thus should be affected by group-level forces. Our findings suggest that group attributes (the social context) can affect the weighing of concerns by individual group members. We also demonstrate how empirical studies can be conducted to explore privacy behaviors at the group level. HLM allowed us to analyze data from individual group members, while also considering group-level effects of group attributes. We note that while HLM has been used in other IS research domains, such as collaborative work (Turel & Zhang, 2011) or phishing (Wright et al., 2023), this is one of the first studies that applied it specifically to information privacy behaviors. We hope this paves the way for HLM applications to address other multilevel research questions.

Last, we use a deceptive paradigm that allows observing actual sharing of co-owned information, as opposed to intentions or willingness to share information. We acknowledge that there have been several recent papers focusing on co-owned information disclosure (Feng et al., 2024, 2023); however, they primarily studied sharing intentions and/or took a self-centric perspective. It is important to study actual behaviors, particularly in the privacy research domain (Buck et al., 2022), because people might say they highly value their privacy, but still share information carelessly (Alashoor et al., 2023; Gerber et al., 2018). Thus, we provide here a paradigm that future research can leverage.

6.2. Practical contributions

The findings have implications for individuals and practitioners. For individuals, our study underscores a potential bias in co-owned information management. We show that in close groups, individuals might under-weigh their own concerns and thus increase their likelihood of sharing co-owned information. Hence, individuals should be educated and informed about this bias and be trained to suppress it. Moreover, while many organizations see value in forming well-bonded teams, they should also be mindful of the dark sides of this practice: it can lead to possibly higher sharing through under-weighing self-focused concerns. Note that we do not suggest that team building should be avoided. Instead, we suggest that in cases where privacy protection is highly important, the benefits of close-knit teams should be weighed against their drawbacks.

For digital platforms, our findings imply that they should help groups maintain privacy. The current privacy settings or protection functions are mostly individually oriented, without taking the characteristics of social groups to which individuals belong into account (Jia & Baumer, 2022). By integrating group-level elements into the design of privacy features, platform designers, or service providers can develop solutions that are better customized to accommodate the needs and privacy preferences of groups of users. As an illustration, when digital platforms detect a high-level closeness group, it could inform members about the bias we observed here and warn against excessive sharing of co-owned information.

Last, for policy makers, a key limitation in current data protection guidelines is the excessive emphasis on personal, individually identifiable information. Despite this emphasis, individual may still be targeted by algorithmic inferential analysis, even without directly engaging in self-disclosure online (Mittelstadt, 2017; Taylor et al., 2016). Therefore, privacy legislation and ethical guidelines could benefit from incorporating protections not only for identifiable individuals but also for group-based and co-owned data that may indirectly reveal sensitive insights about individuals nested within these groups.

6.3. Limitations and future research directions

Several limitations that can inform future research are noteworthy. First, we followed the idea that “context-sensitive theory development requires a deliberate selection of contextual elements” (Sarker, 2016) and considered only one group attribute (i.e., group closeness). However, there are additional factors that might affect the psychological processes of group members. Examples include psychological ownership and people’s AI literacy. We also acknowledge the limitation of not including all group-level factors, which cannot be reasonably done in a single study. Note that our aim is to open the avenue for multilevel privacy research – not to close it – by showcasing how group-level dynamics such as closeness can shape individual privacy behaviors. There are other potential group-level attributes such as group identity, group belongingness, and group salience. We call for future multilevel privacy research to examine such aspects.

Second, we focused on self-centered privacy considerations, a perspective well established in prior research, particularly in the APCO model (Smith et al., 2011). This focus allowed us to build on and extend the literature by showing how individual-level privacy considerations are shaped by group-level factors, highlighting our multilevel contribution. However, in the context of co-owned information, others’ privacy concerns and benefits might also matter. We therefore acknowledge the limitation of not including these others-centered considerations, which presents a valuable direction for future research. We emphasize that our key contribution lies in advancing understanding of how group dynamics influence self-centered privacy decision-making. We encourage future research to explore the complementary or competing role of others-centered privacy considerations in this process, which is an important avenue.

Third, we focused on hedonic benefits, while acknowledging that social and utilitarian benefits can also motivate sharing behaviors. In addition, we focused on two APCO elements: one motivating factor, and another sharing deterrent, without comprehensively examining other APCO constructs such as trust and privacy risks (Dinev et al., 2015). Although these choices represent a reasonable starting point, especially given our study design, they limit eth breadth of our model and its generalizability. We hence call for more research to unpack context effects on the broader range of sharing motivators and deterrents.

Fourth, our study design did not account for all potential alternative explanations, and did not control factors such as whether the picture was actually shared before (even though people self-reported it was unshared). Thus, future research may consider more stringent verification processes. In addition, our study design intentionally excluded co-owner consultation to observe individual decision-making in a controlled environment. However, in real-world settings, decisions regarding co-owned content sharing are often made in consultation with others involved. This limitation should be addressed in future research exploring group-based disclosure dynamics. Future research could explore the role of co-owner consultation in disclosure decisions using qualitative approaches. This could include examining whether such consultations occur, how they are conducted, and what reasons or arguments are exchanged for or against sharing the co-owned content. By capturing these negotiation dynamics, future research can offer deeper insights into the social processes underlying co-owned information sharing in real-life settings.

The fifth limitation of the study design is that participants were recruited through the researchers’ personal networks, which may introduce selection bias or increase the risk of social desirability effects. While initial recruitment relied on the authors’ personal networks, this served mainly to trigger broader participant acquisition via indirect contacts (e.g., friends of friends). While, this was done without revealing the true purpose fo the study, there is always a risk that this approach may have slightly increased participants’ willingness to engage in this study. We believe that this bias is limited due to the indirect recruitment paths, sample diversity and deceptive nature of the invitation. Moreover, our sample primarily consisted of university students, academic staff, and employees, providing access to individuals with varied group affiliations and social ties. While relevant for studying co-owned information sharing—given their active engagement in social networks and online sharing—the sample may not fully represent broader populations. Their sharing behaviors may differ from those of other groups (e.g., older adults, professionals, family contexts). Future research can consider recruiting more diverse populations across social and cultural contexts.

Sixth, we did not include measures of individual differences in privacy-related dispositions, such as sensitivity toward information disclosure. Although our focus was on social relational factors influencing co-owned information sharing, it is conceivable that individual privacy orientations could also affect disclosure decisions. We acknowledge this as a potential limitation and suggest that future research may benefit from examining how such individual dispositions interact with social factors in shaping information-sharing behavior. Moreover, while psychological safety is discussed as a relevant concept in relation to H4, it was not directly measured in the current study. We therefore recommend that future research on group closeness consider incorporating direct assessments of psychological safety to further elucidate its role.

Last, the generalizability of our finding can be established, or boundaries around the generalizability could be set, by considering different types of co-owned information, different formats of shared information (e.g., videos), and different research paradigms (e.g., field experiments). Additionally, we considered two levels: individuals and the groups within which they are nested. It is worth considering additional layers of the social context, such as the culture or organization in which the groups may be nested, i.e., more than two levels. This can further unpack the complex cross-level forces data owners are exposed to.

7. Conclusion

This study took a multilevel approach to theorize and test a two-level model of co-owned information sharing by group members. This model extends models of privacy behaviors by individuals by (1) moving from an individual-level lens to a cross-level lens, and (2) showing that the weighing of privacy concerns can be context-dependent as it is influenced by one's proximal social context, and specifically by the closeness of the groups within which individuals are nested. Ultimately, our insights emphasize the importance of co-owned information management and interdependent privacy protection in today's surveillance economy. We hence call for policy designers and practitioners to focus on co-owned data protection and devise legislation frameworks to protect such data. Overall, our work represents an initial step that can inspire future multilevel privacy research, particularly in group-level theorizing, operationalization, and the application of multilevel analytical methods. We call for more multilevel information privacy research on how group attributes affect privacy decision-making.

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CRediT authorship contribution statement

Anne Zöll: Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ofir Turel:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Mingxin Zhang:** Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Anne Zoell reports financial support was provided by German Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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APPENDIX. : Literature Review about Co-owned Information Disclosure

To gain a comprehensive understanding of extant research on co-owned information disclosure, we performed a brief literature review, described below. We first developed the following search query:

- AB= (data OR information OR privacy) AND AB= (co-owned OR group OR collective OR interdependent OR networked OR multi-party) AND AB= (disclose OR disclosure OR share OR sharing OR provide OR providing)

  We applied the search query to Web of Science and AIS E-library. Our aim was to gather as many relevant papers as possible; hence, we did not have a time span. We also focused on papers published in adjacent disciplines such as Psychology, Social Science Multidisciplinary, Communication Science, Library and Information Science, Organizational Studies, etc. As of 22nd October 2024, the database search yielded 2633 papers in total. We screened each paper by reading its title, abstract, and keywords. A paper would be suitable for further full article assessment if it satisfied the following criteria:
- It was written in English.
- It provided access to its full text.
- It primarily focused on co-owned information disclosure, as opposed to self-disclosure and peer disclosure.

After this stage, twenty-six papers remained. We then scrutinized the collected papers. A paper that met any one of the following criteria would be excluded during the full-article assessment stage:

- It was a technique, meta-analysis, theoretical review, or opinion article.
- It did not involve primary data from human participants.

Finally, we identified eight relevant papers that studied co-owned information disclosure. Despite the potential for some papers to be absent from the search results, we believe our search query and search strategy remain overall effective in identifying the majority of relevant papers.

Table A1
A Synthesis of Co-owned Information Disclosure Research

No.	Citation	Terminology	Predictors	Participants	Methods	Measurement	Level of analysis	Theory	Paper type
1	(Feng et al., 2024)	Co-owned information disclosure	• Personal benefit • Personal risk • Relationship benefit • Relationship risk • Others' benefit • Others' risk • Personal distributive fairness • Others' distributive fairness	740 participants	Cross-sectional surveys	Likert scale	Individual level	• Privacy calculus theory • Justice theory	Full journal article
2	(Feng et al., 2023)	Co-owned information disclosure	• Self-presentation • Others-presentation • Relationship presentation • Personal privacy concerns • Others-privacy concerns • Relationship privacy concerns	740 participants	Cross-sectional surveys	Likert scale	Individual level	• Privacy calculus theory • Impression management theory	Full journal article
3	(Jia & Baumer, 2022)	Collective privacy management	• Relationship support • Relationship risks • Shared values and goals • Group structure • Internal communication • Collective privacy boundary • External communication goals • Privacy concerns • Boundary management	16 interviewees	Interviews	Interview protocol	Individual level	• Communication privacy management theory	Full journal article
4	(Jia & Xu, 2016)	Collective privacy concerns	• Collective information access • Collective information control • Collective information diffusion	427 participants	Cross-sectional surveys	Likert scale	Individual level	• Communication privacy management theory	Full journal article
5	(Such et al., 2017)	Photo sharing	• Sharing prevalence • Severity • Communication before/after sharing • Conflict resolution	1033 participants	Cross-sectional surveys	Likert scale	Individual level	None	Full conference article
6	(Kim & Gweon, 2016)	Picture sharing	• Ownership • Closeness • Group norms	29 participants	Experiments	Likert scale	Individual level	None	Full journal article
7	(Memarian Esfahani & Kim, 2022)	Collective information privacy	• Psychological ownership • Tie strength • Group characteristics	None	None	None	Individual level	• The theory of multilevel information privacy management	Research-in-progress conference paper
8	(Bute & Brann, 2015)	Co-owned private information	• Information co-ownership • Privacy rules	40 interviewees	Interviews	Interview protocol	Individual level	• Communication privacy management theory	Full journal article

References

Acquisti, A. (2009). Nudging privacy: the behavioral economics of personal information. *IEEE Security Privacy, 7*(6), 82–85.

Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behavior in the age of information. *Science, 347*(6221), 509–514.

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*.

Alashoor, T., Keil, M., Smith, H. J., & McConnell, A. R. (2023). Too tired and in too good of a mood to worry about privacy: explaining the privacy paradox through the lens of effort level in information processing. *Information Systems Research, 34*(4), 1415–1436.

Alsarkal, Y., Zhang, N., & Xu, H. (2018, Jan 02-06). Your Privacy Is Your Friend's Privacy: Examining Interdependent Information Disclosure on Online Social Networks. [Proceedings of the 51st annual hawaii international conference on system sciences (hicss)]. 51st Annual Hawaii International Conference on System Sciences (HICSS), Hi.

Altman, I. (1975). The environment and social behavior: privacy, personal space, territory, and crowding.

Appelbaum, N. P., Lockeman, K. S., Orr, S., Huff, T. A., Hogan, C. J., Queen, B. A., & Dow, A. W. (2020). Perceived influence of power distance, psychological safety, and team cohesion on team effectiveness. *Journal of Interprofessional Care, 34*(1), 20–26. https://doi.org/10.1080/13561820.2019.1633290

Bansal, G., Zahedi, F. M., & Gefen, D. (2016). Do context and personality matter? Trust and privacy concerns in disclosing private information online. *Information Management, 53*(1), 1–21. https://doi.org/10.1016/j.im.2015.08.001

Bélanger, F., Cefaratti, M., Carte, T., & Markham, S. E. (2014). Multilevel research in information systems: concepts, strategies, problems, and pitfalls. *Journal of the Association for Information Systems, 15*(9), 1.

Bélanger, F., & Crossler, R. E. (2011). Privacy in the digital age: a review of information privacy research in information systems. *Mis Quarterly*, 1017–1041.

Belanger, F., & James, T. L. (2020). A theory of multilevel information privacy management for the digital era [Article]. *Information Systems Research, 31*(2), 510–536. https://doi.org/10.1287/isre.2019.0900

Benitez, J., Henseler, J., Castillo, A., & Schuberth, F. (2020). How to perform and report an impactful analysis using partial least squares: guidelines for confirmatory and explanatory IS research. Article 103168 *Information Management, 57*(2). https://doi.org/10.1016/j.im.2019.05.003.

Breward, M., Hassanein, K., & Head, M. (2017). Understanding consumers' attitudes toward controversial information technologies: a contextualization approach. *Information Systems Research, 28*(4), 760–774.

Brown, R. (2000). Social identity theory: past achievements, current problems and future challenges. *European Journal of Social Psychology, 30*(6), 745–778. https://doi.org/10.1002/1099-0992(200011/12)30:6<745::Aid-ejsp24>3.0.Co;2-o

Buck, C., Dinev, T., & Anaraky, R. G. (2022). Revisiting APCO. In B. P. Knijnenburg, X. Page, P. Wisniewski, H. R. Lipford, N. Proferes, & J. Romano (Eds.), *Modern Socio-Technical Perspectives on Privacy* (pp. 43–60). Springer International Publishing. https://doi.org/10.1007/978-3-030-82786-1_3.

Bute, J. J., & Brann, M. (2015). Co-ownership of private information in the miscarriage context. *Journal of Applied Communication Research, 43*(1), 23–43.

Cao, Z. K., Hui, K. L., & Xu, H. (2018). An economic analysis of peer disclosure in online social communities. *Information Systems Research, 29*(3), 546–566. https://doi.org/10.1287/isre.2017.0744

Chan, D. (1998). Functional relations among constructs in the same content domain at different levels of analysis: a typology of composition models. *Journal of Applied Psychology, 83*(2), 234.

Cheung, C., Lee, Z. W. Y., & Chan, T. K. H. (2015). Self-disclosure in social networking sites the role of perceived cost, perceived benefits and social influence. *Internet Research, 25*(2), 279–299. https://doi.org/10.1108/IntR-09-2013-0192

Cichy, P., Salge, T. O., & Kohli, R. (2021). Privacy concerns and data sharing in the internet of things: mixed methods evidence from connected cars. *Mis Quarterly, 45*(4).

Clarke, R. (2019). Risks inherent in the digital surveillance economy: a research agenda. *Journal of Information Technology, 34*(1), 59–80.

Coutifaris, C. G., & Grant, A. M. (2022). Taking your team behind the curtain: the effects of leader feedback-sharing and feedback-seeking on team psychological safety. *Organization Science, 33*(4), 1574–1598.

Cushman, F., Young, L., & Hauser, M. (2006). The role of conscious reasoning and intuition in moral judgment: testing three principles of harm. *Psychological Science, 17*(12), 1082–1089. https://doi.org/10.1111/j.1467-9280.2006.01834.x

Dinev, T., & Hart, P. (2006). An extended privacy calculus model for E-commerce transactions. *Information Systems Research, 17*(1), 61–80. https://doi.org/10.1287/isre.1060.0080

Dinev, T., McConnell, A. R., & Smith, H. J. (2015). Research commentary—informing privacy research through information systems, psychology, and behavioral economics: thinking outside the "APCO" box. *Information Systems Research, 26*(4), 639–655.

Ellemers, N., Spears, R., & Doosje, B. (1997). Sticking together or falling apart: In-group identification as a psychological determinant of group commitment versus individual mobility. *Journal of Personality and Social Psychology, 72*(3), 617.

Emami Naeini, P., Degeling, M., Bauer, L., Chow, R., Cranor, L. F., Haghighat, M. R., & Patterson, H. (2018). The influence of friends and experts on privacy decision making in IoT scenarios. In *Proceedings of the ACM on Human-Computer Interaction, 2* pp. 1–26). CSCW.

Erle, T. M., & Topolinski, S. (2017). The grounded nature of psychological Perspective-Taking. *Journal of Personality and Social Psychology, 112*(5), 683–695. https://doi.org/10.1037/pspa0000081

Feng, Y., Sun, Y., Wang, N., & Shen, X.-L. (2024). Co-owned information disclosure and collective privacy calculus on social network platforms: the moderating role of information ownership. *Internet Research*.

Feng, Y., Zhang, Y., & Li, L. (2023). Collective impression management and collective privacy concerns in co-owned information disclosure: the mediating role of relationship support and relationship risk. *Library Hi Tech*.

Gerber, N., Gerber, P., & Volkamer, M. (2018). Explaining the privacy paradox: a systematic review of literature investigating privacy attitude and behavior. *Computers Security, 77*, 226–261.

Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: INDEED a SILVER BULLET. *Journal of Marketing Theory and Practice, 19*(2), 139–151. https://doi.org/10.2753/mtp1069-6679190202

Hall, K., Helmus, B., & Eymann, T. (2024). How to balance privacy and (Health) benefits: privacy calculus and the intention to use health tracking at the workplace. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2024.2375704

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science, 43*, 115–135.

Ho, J. C. F., & Ng, R. (2022). Perspective-Taking of Non-Player characters in prosocial virtual reality games: effects on closeness, empathy, and game immersion. *Behaviour Information Technology, 41*(6), 1185–1198. https://doi.org/10.1080/0144929x.2020.1864018

Hox, J. J., & Kreft, I. G. (1994). Multilevel analysis methods. *Sociological Methods Research, 22*(3), 283–299.

James, L. R., Demaree, R. G., & Wolf, G. (1984). Estimating within-group interrater reliability with and without response bias. *Journal of Applied Psychology, 69*(1), 85.

Jia, H., & Baumer, E. P. (2022). Birds of a feather: collective privacy of online social activist groups. *Computers Security, 115*, Article 102614.

Jia, H., & Xu, H. (2016). Measuring individuals' concerns over collective privacy on social networking sites. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace, 10*(1).

Johns, G. (2006). The essential impact of context on organizational behavior. *Academy of Management Review, 31*(2), 386–408.

Jones, K., & Duncan, C. (1998). Modelling context and heterogeneity: applying multilevel models. *Research strategies in the social sciences: a guide to new approaches* (pp. 95–125). Oxford University Press.

Jowett, S., Do Nascimento-Júnior, J. R. A., Zhao, C., & Gosai, J. (2023). Creating the conditions for psychological safety and its impact on quality coach-athlete relationships. *Psychology of Sport and Exercise, 65*, Article 102363.

Junco, R. (2013). Comparing actual and self-reported measures of facebook use. *Computers in Human Behavior, 29*(3), 626–631.

Karwatzki, S., Trenz, M., & Veit, D. (2022). The multidimensional nature of privacy risks: conceptualisation, measurement and implications for digital services. *Information Systems Journal, 32*(6), 1126–1157. https://doi.org/10.1111/isj.12386

Kawakami, K., Amodio, D. M., & Hugenberg, K. (2017). Intergroup perception and cognition: an integrative framework for understanding the causes and consequences of social categorizationJ. M. Olson (Ed.). *Advances in Experimental Social Psychology, Vol 55, 55*, 1–80. https://doi.org/10.1016/bs.aesp.2016.10.001

Kehr, F., Kowatsch, T., Wentzel, D., & Fleisch, E. (2015). Blissfully ignorant: the effects of general privacy concerns, general institutional trust, and affect in the privacy calculus. *Information Systems Journal, 25*(6), 607–635. https://doi.org/10.1111/isj.12062

Keith, M. J., Thompson, S. C., Hale, J., Lowry, P. B., & Greer, C. (2013). Information disclosure on mobile devices: Re-examining privacy calculus with actual user behavior. *International Journal of Human-Computer Studies, 71*(12), 1163–1173. https://doi.org/10.1016/j.ijhcs.2013.08.016

Khabbache, H., Ouazizi, K., Bragazzi, N. L., Alaoui, Z. B., & Mrabet, R. (2020). Thinking about what the others are thinking about: an integrative approach to the mind perception and social cognition theory. *Cosmos and History-the Journal of Natural and Social Philosophy, 16*(2), 266–295. <Go to ISI>://WOS:000581598100011.

Kim, A., & Gweon, G. (2016). Comfortable with friends sharing your picture on facebook? - effects of closeness and ownership on picture sharing preference. *Computers in Human Behavior, 62*, 666–675. https://doi.org/10.1016/j.chb.2016.04.036

Klein, K. J., Tosi, H., & Cannella, A. A., Jr (1999). Multilevel theory building: benefits, barriers, and new developments. *Academy of Management Review, 24*(2), 248–253.

Koohikamali, M., Peak, D. A., & Prybutok, V. R. (2017). Beyond self-disclosure: disclosure of information about others in social network sites. *Computers in Human Behavior, 69*, 29–42. https://doi.org/10.1016/j.chb.2016.12.012

Kozlowski, S. W., Chao, G. T., Grand, J. A., Braun, M. T., & Kuljanin, G. (2013). Advancing multilevel research design: capturing the dynamics of emergence. *Organizational Research Methods, 16*(4), 581–615.

Krasnova, H., Veltri, N. F., & Günther, O. (2012). Self-disclosure and privacy calculus on social networking sites: the role of culture - intercultural dynamics of privacy calculus. *Business Information Systems Engineering, 4*(3), 127–135.

Li, H., Luo, X., Sarathy, R., & Xu, H. (2024). Contextual Integrity and Personal Health Information Sharing on Online Social Networks.

Li, H., Sarathy, R., & Xu, H. (2011). The role of affect and cognition on online consumers' decision to disclose personal information to unfamiliar online vendors. *Decision Support Systems, 51*(3), 434–445.

Li, H., Sarathy, R., & Zhang, J. (2008). The role of emotions in shaping consumers' privacy beliefs about unfamiliar online vendors. *Journal of Information Privacy and Security, 4*(3), 36–62.
Li, Y., & Gui, X. (2022). Examining co-owners' privacy consideration in collaborative photo sharing. *Computer Supported Cooperative Work (CSCW), 31*(1), 79–109.
Magni, M., Proserpio, L., Hoegl, M., & Provera, B. (2009). The role of team behavioral integration and cohesion in shaping individual improvisation. *Research Policy, 38*(6), 1044–1053.
Memarian Esfahani, S., & Kim, D.J. (2022). The collective information privacy on social networking sites.
Min, J., & Kim, B. (2015). How are people enticed to disclose personal information despite privacy concerns in social network sites? The calculus between benefit and cost. *Journal of the Association for Information Science and Technology, 66*(4), 839–857. https://doi.org/10.1002/asi.23206
Mittelstadt, B. (2017). From individual to group privacy in big data analytics. *Philosophy Technology, 30*(4), 475–494. https://doi.org/10.1007/s13347-017-0253-7
Molina-Azorín, J. F., Pereira-Moliner, J., López-Gamero, M. D., Pertusa-Ortega, E. M., & José Tarí, J. (2020). Multilevel research: foundations and opportunities in management. *BRQ Business Research Quarterly, 23*(4), 319–333.
Nissenbaum, H. (2004). Privacy as contextual integrity. *Washington Law Review, 79*, 119.
Oztop, P., Katsikopoulos, K., & Gummerum, M. (2018). Creativity through connectedness: the role of closeness and perspective taking in group creativity. *Creativity Research Journal, 30*(3), 266–275.
Peng, Z. (2023). Your growth is my growth: examining sharenting behaviours from a multiparty privacy perspective. *Communication Research and Practice, 9*(3), 271–289. https://doi.org/10.1080/22041451.2023.2216584
Peterson, J. L., Bellows, A., & Peterson, S. (2015). Promoting connection: Perspective-taking improves relationship closeness and perceived regard in participants with low implicit self-esteem. *Journal of Experimental Social Psychology, 56*, 160–164. https://doi.org/10.1016/j.jesp.2014.09.013
Petronio, S. (2010). Communication privacy management theory: what do we know about family privacy regulation? *Journal of Family Theory Review, 2*(3), 175–196.
Polman, E., & Wu, K. Y. (2020). Decision making for others involving risk: a review and meta-analysis. Article 102184 *Journal of Economic Psychology, 77*. https://doi.org/10.1016/j.joep.2019.06.007.
Porter, J., McDermott, T., Daniels, H., & Ingram, J. (2024). Feeling part of the school and feeling safe: further development of a tool for investigating school belonging. *Educational Studies, 50*(3), 382–398.
Prinstein, M. J., & Wang, S. S. (2005). False consensus and adolescent peer contagion: examining discrepancies between perceptions and actual reported levels of friends' deviant and health risk behaviors. *Journal of Abnormal Child Psychology, 33*, 293–306.
Saffarizadeh, K., Keil, M., Boodraj, M., & Alashoor, T. (2024). "My name is alexa. What's Your Name?" the impact of reciprocal Self-Disclosure on Post-Interaction trust in conversational agents. *Journal of the Association for Information Systems, 25*(3), 528–568.
Sagan, A. (2019). Sample size in multilevel structural equation modeling–the Monte Carlo approach. *Econometrics, 23*(4), 63–79.
Sanner, B., & Bunderson, J. S. (2015). When feeling safe isn't enough: contextualizing models of safety and learning in teams. *Organizational Psychology Review, 5*(3), 224–243.
Sarker, S. (2016). Building on davison and martinsons' concerns: a call for balance between contextual specificity and generality in IS research. *Journal of Information Technology, 31*(3), 250–253.
Simmering, M. J., Fuller, C. M., Richardson, H. A., Ocal, Y., & Atinc, G. M. (2015). Marker variable choice, reporting, and interpretation in the detection of common method variance: a review and demonstration. *Organizational Research Methods, 18*(3), 473–511.
Smith, H. J., Dinev, T., & Xu, H. (2011). Information privacy research: an interdisciplinary review. *Mis Quarterly*, 989–1015.
Such, J. M., Porter, J., Preibusch, S., & Joinson, A. (2017). Photo privacy conflicts in social media: a large-scale empirical study. *Proceedings of the 2017 CHI Conference on Human factors in Computing Systems*.
Sun, Y. Q., Wang, N., Shen, X. L., & Zhang, J. X. (2015). Location information disclosure in location-based social network services: privacy calculus, benefit structure, and gender differences. *Computers in Human Behavior, 52*, 278–292. https://doi.org/10.1016/j.chb.2015.06.006
Tamir, D. I., & Mitchell, J. P. (2012). Disclosing information about the self is intrinsically rewarding. *Proceedings of the National Academy of Sciences, 109*(21), 8038–8043.
Taylor, L., Floridi, L., & Van der Sloot, B. (2016). *Group privacy: New challenges of data technologies, 126*. Springer.
Tierney, M., Spiro, I., Bregler, C., & Subramanian, L. (2013). Cryptagram: photo privacy for online social media. *Proceedings of the First ACM Conference on Online Social Networks*.
Turel, O., & Zhang, Y. J. (2011). Should I e-collaborate with this group? A multilevel model of usage intentions. *Information Management, 48*(1), 62–68.
Utz, S., & Beukeboom, C. J. (2011). The role of social network sites in romantic relationships: effects on jealousy and relationship happiness. *Journal of Computer-Mediated Communication, 16*(4), 511–527.
Voss, K. E., Spangenberg, E. R., & Grohmann, B. (2003). Measuring the hedonic and utilitarian dimensions of consumer attitude. *Journal of Marketing Research, 40*(3), 310–320.
Wisniewski, P. J., & Page, X. (2022). Privacy theories and frameworks. In B. P. Knijnenburg, X. Page, P. Wisniewski, H. R. Lipford, N. Proferes, & J. Romano (Eds.), *Modern Socio-Technical Perspectives on Privacy* (pp. 15–41). Springer International Publishing. https://doi.org/10.1007/978-3-030-82786-1_2.
Wray, L. D., & Stone, E. R. (2005). The role of self-esteem and anxiety in decision making for self versus others in relationships. *Journal of Behavioral Decision Making, 18*(2), 125–144. https://doi.org/10.1002/bdm.490
Wright, R. T., Johnson, S. L., & Kitchens, B. (2023). Phishing susceptibility in context: A multilevel information processing perspective on deception detection. *MIS Quarterly, 47*(2).
Xu, C., Gao, F., & Han, L. (2025). Enhancing user information disclosure intention in dynamic conversations of intelligent recommendation systems based on large language models: a perspective of user gratification and privacy calculus. *International Journal of Human-Computer Studies, 200*, Article 103511.
Xu, H., Teo, H. H., Tan, B. C. Y., & Agarwal, R. (2009). The role of Push-Pull technology in privacy calculus: the case of Location-Based services. *Journal of Management Information Systems, 26*(3), 135–173. https://doi.org/10.2753/mis0742-1222260305
Xu, H., & Zhang, N. (2022). From contextualizing to context theorizing: assessing context effects in privacy research. *Management Science, 68*(10), 7383–7401.
Zalmanson, L., Oestreicher-Singer, G., & Ecker, Y. (2022). The role of social cues and trust in users? private information disclosure. *MIS Quarterly, 46*(2), 1109–1134. https://doi.org/10.25300/misq/2022/16288
Zhang, N., Wang, C., Karahanna, E., & Xu, Y. (2022). Peer privacy concern: Conceptualization and measurement. *MIS Quarterly, 46*(1), 491–530. https://doi.org/10.25300/misq/2022/14861